



Sizhang Xu

Full-Stack Engineering · Real-Time Systems

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EDUCATION

University of Washington

Expected Jun 2028

M.S. in Technology Innovation · Global Innovation Exchange (GIX) · In progress

Tsinghua University

Graduate studies · GIX dual-degree program

Lab affiliation: PI Lab, Department of Computer Science and Technology

Shanghai Jiao Tong University

Jun 2025

B.Eng. in Industrial Design · Rank 1/30 · National Scholarship

Semester exchange: University College London

INTERNSHIP EXPERIENCE

ByteDance | Seed

Feb - Jun 2026

LLM Data Product Intern | **LangGraph** Evaluation Agent

- **Product context:** Led product development of DataSeek's post-evaluation analysis module from concept through iterative releases, translating quantitative model-evaluation results into capability-level failure diagnoses and training-data recommendations. Owned PRDs, prompt architecture, and workflow design; developed and iterated a **Python/LangGraph** prototype and coordinated implementation with engineering.
- **Analysis workflow:** Designed an agent pipeline for reference reasoning → comparative error diagnosis → capability aggregation → training-data mapping. Used **best-of-N** reference sampling and LLM judging to compare evaluated-model and reference reasoning, extract question-level failure labels, and aggregate fine-grained findings into capabilities mapped to pretraining, SFT, and RL data needs. Designed a **four-quadrant analysis framework** (reference answer present/absent × scoring rubric present/absent) to systematically handle objective and subjective evaluation tasks.
- **Prompt engineering:** Led the design of **30+** structured prompt templates, establishing scenario-action capability naming and constraints against unsupported capability attribution. To address diluted long-tail failures and spurious capability labels, redesigned aggregation into two stages: bucket by predefined capability labels, then group failures within each bucket to sharpen diagnoses and support cross-dataset comparison.
- **Product design and iteration:** Independently authored PRDs and coordinated multiple review rounds. Led manual quality reviews of **700+** reports, triaged nearly **100 issues** by **P0/P1/P2** priority, and established a closed-loop process from issue identification to requirements and fix tracking. Designed a dual-track improvement plan: short-term frontend safeguards and long-term prompt-engineering optimization.

Sohu | Mint Lab

Oct 2025 - Jan 2026

AI Product Management Intern

- **Product context:** Developed an AI-assisted creative-content tool for creators facing information overload, fragmented ideas, and difficulty organizing material. Structured the workflow around cognitive steps and the creative process, turning divergent ideas into editable, traceable topic graphs and drafts.
- **Prompt architecture:** Led prompt design across association → structuring → linear overview → drafting. Introduced **Root/Hyponym** hierarchy constraints and chain-of-thought prompting, combined with self-ranking and voting, to address uncontrolled topic expansion, hallucinations, and inconsistent parent-child relationships.
- **Engineering and iteration:** Independently developed Python end-to-end test scripts. Split title and description generation into sequential model calls to refine the execution pipeline; designed an Outline-to-Draft generation flow for granular editing and traceability.
- **Evaluation framework:** Defined a **six-dimension** quality rubric with stage-specific weights and built a four-level evaluation set from recurring content topics. Selected model combinations under quality, latency, and cost constraints, using regression cases to quantitatively validate prompt iterations and check stability.

INTERNSHIP EXPERIENCE · CONTINUED

Tsinghua University | Future Laboratory

Jun - Sep 2024

Human-Computer Interaction Research Intern

- **System modeling and prototyping:** Built a wearable force-feedback prototype and modeled the relationship between motor input angle and output resistance. Implemented three **C#/Unity** modules for cutting fruit and vegetables, chopping a wooden target, and tearing paper.
- **Testing and iteration:** Integrated virtual contact events with physical resistance and iterated interaction timing. Evaluated the prototype with **eight participants**, using observations and signal traces to investigate timing mismatches and compensatory effort near the end of a cutting motion.

PUBLICATIONS

"In Sync with Me": A Robotic Body-Doubling Probe for ADHD Programmers' Deep Work

Chenyang Li, **Sizhang Xu**, Hancheng Ye, and Ziyu Huang

UbiComp Companion '26 (Poster), 2026 · Accepted / in press

[DOI](#)

SELECTED PROJECTS

Real-Time Sleep Research Platform

2026

Python · **FastAPI** · Vue/JavaScript · **WebSocket** · BLE

- Implemented **FastAPI** REST controls and **WebSocket** runtime snapshots for connection setup, live signal inspection, intervention status, and event timelines. Exposed disconnected devices, stale readings, and cooldown conditions as explicit interface states.
- Built a Python research platform connecting BLE sensors, audio inference, intervention control, and a Vue interface. Separated hardware access, control logic, and recorded-session review so researchers could configure and inspect an experiment in one application.
- Separated audio capture, pretrained **TensorFlow/YAMNet** inference, and BLE I/O with a **ring buffer**, an inference worker, and a dedicated **asyncio** event loop. Kept model inference out of the audio-capture callback and used configurable thresholds and state smoothing.
- Implemented a trigger, playback, observation, and cooldown **state machine**. Recorded timestamped events in **JSONL** and moved buffered waveform compression and **NPZ** block writes to a background thread for session review.

Alpha-TradingAgents | Multi-Agent Equity Research

Summer 2026

Python · **LangGraph** · akshare | github.com/xuxu8421/alpha-trading-agents

- Extended the open-source TradingAgents framework for A-share equity research, adding market, financial, news, and policy data adapters. Implemented **exponential-backoff** retries around data requests and connected the resulting inputs to the existing analyst workflow.
- Added custom policy-analysis and report-writer nodes to **LangGraph**. Combined analyst evidence with the framework's bull/bear debate and risk-review outputs to produce a synthesized research report. The extension reused the framework's existing orchestration.
- Defined a structured metadata interface between report synthesis and PDF rendering, separating research logic from presentation. Built the report renderer to populate summary cards from those fields and generated a **four-page** research-report sample.

CAMPUS LEADERSHIP

SJTU Student Union — Presidium Member

School of Design Student Union, SJTU — Executive President

Nanyang News Agency, SJTU — Presidium Member & Deputy Secretary-General

SKILLS

Languages: Python, JavaScript, C#, HTML/CSS

Engineering: **LangGraph**, **FastAPI**, Vue, REST APIs, **WebSocket**, **asyncio**, Git

Evaluation & product: LLM evaluation, prompt design, regression testing, requirements, Figma